

Evanns Morales-Cuadrado

Robotics PhD Candidate | U.S. Citizen | Fulbright Specialist

CONTACT

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SUMMARY

Robotics PhD candidate in Dr. Samuel Coogan's FACTS Lab at Georgia Tech and AI for Autonomy intern at Sandia National Laboratories. Specializes in safe autonomy, reinforcement learning, and real-time control. Proven track record of developing and deploying safe, computationally efficient planning and control methods for autonomous systems, with work published in leading peer-reviewed robotics and controls venues. Selected to the U.S. Department of State's Fulbright Specialist Roster after spearheading an international research collaboration with a Brazilian partner institution and securing research funding.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology

2022-Present

Ph.D. in Robotics

Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington

2018-2022

Honors Bachelor of Science in Electrical Engineering

Minor in Mathematics

Minor in Physics

Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Year-Round Graduate Research Intern | AI for Autonomy

May 2026 - Present

Sandia National Laboratories

Albuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

Graduate Research Assistant | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight tracking control law that beat PX4's native tracker in aggressive quadrotor flight and matched NMPC's tracking accuracy at far lower computation time and CPU energy
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

Research Assistant | DN&C and Autonomous Systems Labs

October 2019 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Frank Lewis in the Autonomous Systems Lab and Dr. Yan Wan in the Dynamical Networks and Control Lab
- Trained in one of the most-cited control research groups in electrical engineering, with Dr. Lewis' work ranked 12th in the USA and 23rd worldwide by citation impact
- Contributed to reinforcement learning-based quadrotor control research
- Completed UTA Honors College thesis on learning-based minimum-time and minimum-energy multi-vehicle path planning

SELECTED PUBLICATIONS

E. Morales-Cuadrado, L. Chung, S. Kousik, and S. Coogan, "RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 MECC-ALDSC. (*accepted to both*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

E. Morales-Cuadrado, L. Baird, Y. Wardi, and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." [IEEE Transactions on Control Systems Technology \(TCST\)](#), 2026. [preprint available](#).

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot**: Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4**: Developed and open-sourced a ROS 2 C++ Vicon-PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech's Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

Co-Founder and President of Puerto Rican Student Association

August 2024 - Present

Georgia Institute of Technology

Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

HONORS AND AWARDS

Fulbright Specialist

August 2026

Peer-review-selected to the U.S. Department of State's Fulbright Specialist Roster for a three-year term

Top-3 Finalist at a Deep Learning Research Symposium

November 2023

Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech

August 2022

Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington

May 2022

Exceptional academic distinction in top GPA tier for graduation honors

Electrical Engineering Honors Scholar at University of Texas at Arlington

December 2021

Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership

August 2021

Selective undergraduate society recognizing outstanding academic achievement and community service